

Tutorial 1 Notes - RA G10+G11

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1. The Peano Axioms :

The Peano Axioms are a set of axioms given by Giuseppe Peano, and **used to define the arithmetical properties of natural numbers**, represented as the set $\mathbb{N}$. Using them, the set is primarily defined as:

1. **It contains the number 0.** You may have been taught previously that 0 isn't a natural number, but under the Peano axioms (which is what we will use in this course), 0 is a natural number.
2. **It contains a unary (i.e. using one operand) operation, known as the “successor” operation, S , where $S(n) = n++$.** You can visualise this operation as $S(n) = n++ =$ the element after n in the set $\mathbb{N}$. Later, you can also prove that $n++ = n + 1$, but we will not use that definition right now, and will stick to $n++$.

Using these definitions, we can properly state the Peano Axioms, which are:

P1. $0 \in \mathbb{N}$.

P2. if $n \in \mathbb{N}$, then $n++ \in \mathbb{N}$.

P3. 0 is not the successor of any natural number, i.e. $n++ \neq 0$, for all $n \in \mathbb{N}$.

NOTE: The statements “0 is not a successor” and “Every nonzero number is a successor” are **NOT** the same. While P3 directly asserts the first statement, it **does not** assert the second one, i.e. you **cannot directly** use the second statement from the Peano axioms, you will have to prove it first. The proof for the statement “every nonzero number is a successor” can be proven from P3 using simple induction.

P4. if $n++ = m++$, then $n = m$.

P5. This axiom is also called the **Principle of Mathematical Induction**. It states that if $P(0) = \text{true}$, and if when $P(n) = \text{true}$, $P(n++)$ is true, then P is true for all $n \in \mathbb{N}$. Here, P is any property of natural numbers $n \in \mathbb{N}$.

The principle of induction gives us a way to prove that a property $P(n)$ is true for every natural number n .

NOTE : Because this axiom refers not just to variables, but also properties, it is of a different nature than the other four axioms, and can be called an axiom *schema* rather than an axiom, since it serves as a template for producing an infinite number of axioms, rather than being a single axiom itself. However, for the purposes of this course, we will continue to call it an axiom, i.e. the fifth Peano Axiom P5.

The **informal** intuition behind this axiom is the following. Suppose $P(n)$ is such that $P(0)$ is true, and such that whenever $P(n)$ is true, then $P(n++)$ is true. Then, since $P(0)$ is true, $P(0++) = P(1)$ is true. Since $P(1)$ is true, $P(1++) = P(2)$ is true. Repeating this indefinitely, we can conclude that $P(0), P(1), P(2), P(3)$, etc. are all true. - **(1)**

We cannot, however, conclude through this line of reasoning that, for example, $P(0.5)$ is true. Thus Axiom 2.5 should not necessarily always hold for number systems which contain “unnecessary” elements, such as 0.5.

We can give a “proof” (this proof is **very informal and not valid yet** since we have not defined notions as “integer”, “half-integer”, and “0.5”, it is **only helpful to give you a conceptual understanding of induction**) of this fact.

Take the property $P(n) = “n \text{ is not a half-integer (i.e., an integer plus } 0.5)”$, and apply induction. So, $P(0)$ is true (as 0 is not a half-integer), and if $P(n)$ is true, then $P(n++)$ is true.

Axiom 2.5 asserts that $P(n)$ is true for all natural numbers n (from the line **(1)**), therefore, no natural number can be a half-integer.

So, 0.5 cannot be a natural number.

Since we defined Peano Axioms for natural numbers, you can say that it wouldn’t hold for numbers such as 0.5. This “proof” can be repeated for any such “unnecessary”, non-natural number, and you can get an idea as to how the principle of induction prohibits any numbers other than the “true” natural numbers from appearing in $\mathbb{N}$.

2. Strict Linear Order, Linearly Ordered Sets and Least Elements :

Let us take a binary (i.e. using two operands) relation, $<$, where **$a < b$ means that elements a and b were compared on some factor (this factor can be anything on which comparison is possible), and placed in that manner.** To understand what saying $a < b$ means in a more tangible sense, we would have to know the factor on which the elements were compared which, at this point, we do not. So, we will assume the above definition. We also have **$a = b$, where a and b were compared on a factor and were found to be equal.**

Definition. : Let M be a set. We say $(M, <)$ is a **linearly ordered set**, and $<$ is a **strict linear order** on M , if for all $x, y, z \in M$, the following three statements hold:

1. **$x < x$ is false. $\Rightarrow$ Irreflexivity**
2. **If $x < y$ and $y < z$, then $x < z$. $\Rightarrow$ Transitivity**
3. **Exactly one of $x < y$, $x = y$, $y < x$ holds. $\Rightarrow$ Trichotomy**

If the above relation holds, we can also say $x \leq y$ is “ $x < y$ or $x = y$ ”, and $y > x$ is the same as $x < y$. The property of Trichotomy also gives us the property of ‘**Anti-Symmetry**’, i.e. if $x \leq y$ and $y \leq x$, then $x = y$.

NOTE: When we enforce the above three rules, we can see that the elements follow a linear order, i.e. form a sort of line, where one element either is before ($<$), after ($>$), or the same as ($=$) another element. No element falls outside this line, thus the set is a **linearly ordered set**.

Definition. : Let $(M, <)$ be a linearly ordered set, and let $A \subseteq M$. An element m is a **least element** of A if

$$m \in A \text{ and } m \leq a \text{ for every } a \in A.$$

Proposition. (Uniqueness). *A subset A of a linearly ordered set has at most one least element.*

Proof. Let m and m' both be least elements of A . Since m and m' are both least elements, they must also be elements of A .

Since m is a least element and $m' \in A$, we have $m \leq m'$ by definition. Similarly, since m' is a least element and $m \in A$, we have $m' \leq m$.

That is, $m < m'$ or $m = m'$, and $m' < m$ or $m = m'$ (since $x \leq y$ is “ $x < y$ or $x = y$ ”). **By trichotomy, only one of $m < m'$, $m = m'$, $m' < m$ can hold**, so both requirements can only be satisfied simultaneously if $m = m'$.

Hence, if m and m' are both least elements of A , then $m = m'$.

3. The Well-Ordering Principle

Definition. : Let $(M, <)$ be a linearly ordered set. We say $(M, <)$ is **well-ordered**, or that it satisfies the **well-ordering principle**, if every nonempty subset of M has a least element. Formally,

$$\forall A[(A \subseteq M \wedge A \neq \emptyset) \rightarrow \exists m \in A, \forall a \in A, (m \leq a)].$$

Let us look at some **instances (not entire sets)** where the above principle works, and where it fails, and why:

1. $\mathbb{N}$, **set of natural numbers**, is non-empty, and has least element 0.
2. $A = \{n \in \mathbb{N} : n \neq 0\}$, is non-empty, and has least element 0++.
3. $A = \{n \in \mathbb{N} : P(n) \text{ is false}\}$ i.e. **set of all n for which $P(n)$ is false**, is non-empty for any property P that fails somewhere, and has a least element, which is the least value at which P fails.
4. **A descending chain set** $D = \{d_0, d_1, d_2, \dots\}$, where $d_0 > d_1 > d_2 > \dots$, the set is non-empty, but has **no** least element. For any d_k that we set as least element, there is a $d_{k+1} < d_k$, which means d_k cannot be the least element. So, we cannot get a least element for this set.

5. $M = \{b\} \cup D$, D being the set from the previous example, where we take b such that $b < d_i$ for every i . The set is non-empty, and although $D \subseteq M$ is bounded below (by b , since $b < \text{all elements of } D$), D does not have a least element, since $b \notin D$, and cannot be its least element.

NOTE: Thus, we can see that a set being bounded below by an element is NOT the same as that set having a least element, since the element it is being bounded by is not necessarily in the set itself.

Here, the set B isn't well-ordered either, since D is a subset of B without a least element.

4. Principle of Mathematical Induction and Principle of Well-Ordering

Let us try proving a few statements to derive the relationship between the Principles of Mathematical Induction (or simply, induction) and of Well-Ordering :

First, we will try to imply induction from well-ordering :

Statement. Assume $P1-P4$ and the well-ordering principle, but not $P5$. Derive the principle of mathematical induction.

Proof. Let P be a property such that $P(0)$ is true and such that $P(n)$ implies $P(n++)$ for every n . To prove the principle of mathematical induction, we have to now prove that this means $P(n)$ is true $\forall n \in \mathbb{N}$. Put

$$A = \{n \in \mathbb{N} : P(n) \text{ is false}\}$$

If $A = \emptyset$, there is no n for which $P(n)$ is false, i.e. $P(n)$ is true $\forall n$, which is what we want to prove. Let us use proof by contradiction, and suppose $A \neq \emptyset$. By the well-ordering principle, A has a least element m .

Step 1. Since $P(0)$ is true, $0 \notin A$; as $m \in A$, we get $m \neq 0$.

Step 2. Since $m \neq 0$, we may write $m = p++$ for some $p \in \mathbb{N}$.

Step 3. Since $p++ = m$ and $p < p++$, we can say $p < m$ and since m is least in A , we have $p \notin A$, so $P(p)$ is true.

Step 4. By our hypothesis, since $P(p)$ is true, $P(p++)$ is true, that is, $P(m)$ is true, so $m \notin A$.

Steps 1 and 4 contradict each other, which means A cannot have a least element. Since we assumed well-ordering to be true, A must be $\emptyset$, that is, $P(n)$ holds for every $n \in \mathbb{N}$.

We see some assumptions being made in the above proof, i.e. that **since m is nonzero, it is a successor**, and that **$p < p++$** . Both of these are not axioms, but are **theorems that are either proven directly or use concepts that are proven directly using induction**. Therefore, they **cannot** be used to prove induction itself, since that makes the proof circular and not well-formed. Hence, the above proof becomes an **invalid** proof for induction.

A more fundamental problem is that **to assume well-ordering, we need to define the $\leq$ relation, which in turn needs to be constructed through addition and recursion, which are proven through induction.** This means that **the well-ordering principle isn't a well-formed statement if we only consider the basic Peano Axioms P1-P4.**

We have to make some modifications to the statement and take some base assumptions so that we get a valid proof. To assume well-ordering principle, we need to make the relation primitive i.e. define it on Peano's Axioms itself, and we need to directly assume that the earlier theorems are valid. Therefore, we get a new statement :

Statement. Assume the following, along with P1-P4:

(A) $\leq$ is a primitive relation on $\mathbb{N}$ which is a strict linear order on $\mathbb{N}$ as defined earlier; that is, it satisfies irreflexivity, transitivity, and trichotomy. As before, $x \leq y$ means " $x < y$ or $x = y$ ".

(W) (Well-ordering.) Every nonempty subset of $\mathbb{N}$ has a least element with respect to $<$.

(S) (Predecessors exist.) Every $n \neq 0$ is a successor: there is $p \in \mathbb{N}$ with $p++ = n$.

(O) (Successors increase.) $n < n++$ for every $n \in \mathbb{N}$.

Derive the principle of mathematical induction in the form of P5: if $P(0)$ is true and $P(n)$ implies $P(n++)$ for every n , then $P(n)$ is true for every n .

Proof. Let P be a property of natural numbers such that

(a) $P(0)$ is true, and

(b) for every $n \in \mathbb{N}$, if $P(n)$ is true then $P(n++)$ is true.

Define the set of counterexamples i.e. the set where $P(n)$ is *not* true -

$$A = \{n \in \mathbb{N} : P(n) \text{ is false}\}.$$

Our aim is to show $A = \emptyset$. Suppose instead that $A \neq \emptyset$.

Step 1 (a minimal counterexample exists). By (W), the set A has a least element m ; that is, $m \in A$ and $m \leq a$ for every $a \in A$.

Step 2 (the minimal counterexample is not 0). By hypothesis (a), $P(0)$ is true, so $0 \notin A$. Since $m \in A$, we conclude $m \neq 0$.

Step 3 (it has a predecessor). By (S), since $m \neq 0$, there exists $p \in \mathbb{N}$ with $p++ = m$.

Step 4 (the predecessor is strictly smaller). By (O) applied to p , we have $p < p++$, and $p++ = m$; hence $p < m$.

Step 5 (the predecessor is not a counterexample). Suppose, for contradiction, that $p \in A$. Since m is a lower bound for A , this gives $m \leq p$, so either $m = p$ or $m < p$.

i If $m = p$, then Step 4 reads $p < p$, contradicting irreflexivity in (A).

If $m < p$, then combining with $p < m$ from Step 4 and applying transitivity in (A) yields $m < m$, again contradicting irreflexivity.

Both cases are impossible, so $p \notin A$.

ii Alternatively, we can use trichotomy from (A) and say that since we have said $p < m$, neither $m < p$ nor $m = p$ can be true, so $m \leq p$ cannot be true, i.e. $p \notin A$.

Either way, we get that $P(p)$ must be true.

Step 6 (contradiction). Since $P(p)$ is true, hypothesis (b) gives that $P(p++)$ is true. But $p++ = m$, so $P(m)$ is true, that is, $m \notin A$. This contradicts $m \in A$ from Step 1.

Conclusion. The assumption $A \neq \emptyset$ is false. Therefore $A = \emptyset$, which says precisely that $P(n)$ is true for every $n \in \mathbb{N}$.

From the above statements, we can see that **we cannot directly imply induction from well-ordering, we have to take a few other assumptions alongside well-ordering in order to do that.**

NOTE : Before the next statement, let us state some facts that we get from establishing order in $\mathbb{N}$ (i.e. from saying that $\mathbb{N}$ is linearly ordered). From the Peano Axioms and using induction, we can prove three facts :

F1. $0 \leq n$ for every $n \in \mathbb{N}$.

F2. For all $a, n \in \mathbb{N}$: $n < a$ if and only if $n++ \leq a$.

F3. For all $a, b \in \mathbb{N}$, exactly one of $a < b$, $a = b$, $b < a$ holds. (Trichotomy)

Now, let us try to imply well-ordering from induction :

Statement. Assume the Peano axioms P1-P5, together with the order facts (F1)-(F3). Prove the well-ordering principle for $\mathbb{N}$: every nonempty subset of $\mathbb{N}$ has a least element.

Proof. Let $A \subseteq \mathbb{N}$ be nonempty. We argue by contradiction: suppose that A has no least element.

As we have defined the least element as an element $m \in A$ such that $m \leq a, \forall a \in A$, in order to fully contradict the statement “an element n is the least element of A ”, we must say that there is *no element* $a \in A$ such that $a \leq n$.

So, consider the property

$$P(n) : \text{no element of } A \text{ is } \leq n.$$

If $P(n)$ is true, n is not the least element of A .

By proving that $P(n)$ holds for every n , we can say that every $n \in A$ is not its least element, i.e. A has no least element.

We will prove this using Axiom P5, i.e. induction, since we have taken P5 to be true in this statement.

Base case :

We claim $P(0)$ is true. We shall prove this by contradiction

Step 1. Suppose that some $a \in A$ satisfies $a \leq 0$.

Step 2. By (F1) we also have $0 \leq a$, and hence $a = 0$ (by anti-symmetry); thus $0 \in A$, since we already said that $a \in A$.

Step 3. Since (F1) says that $0 \leq x$ for every $x \in \mathbb{N}$, 0 is a lower bound for A .

Step 4. So, we get $0 \in A$ and $0 \leq a, \forall a \in A$, i.e. 0 is the least element of A , contrary to our supposition that A has no least element.

Therefore our supposition in Step 1 is wrong, and no element of A can be ≤ 0 , which means $P(0)$ is true. We have now proven the base case.

Inductive step :

Assume $P(n)$ is true: no element of A is $\leq n$. We claim $P(n++)$ is true. We will prove this using proof by contradiction.

Step 1. Suppose that some $a \in A$ satisfies $a \leq n++$.

Step 2. Since $P(n)$ holds and $a \in A$, we have $a \not\leq n$; by trichotomy (F3, since either $a < n$, $a = n$ or $a > n$ is true, and $a \not\leq n$ means $a < n$ and $a = n$ is not true, therefore), $n < a$ must be true.

Step 3. By (F2), $n < a$ yields $n++ \leq a$.

Step 4. Combining with $a \leq n++$ gives $a = n++$ (again, using anti-symmetry). Hence $n++ \in A$, as we know $a \in A$.

Step 5. Let $x \in A$ be arbitrary. By $P(n)$ we have $x \not\leq n$, so $n < x$ by (F3), so $n++ \leq x$ by (F2). Thus $n++$ is a lower bound for A .

So, $n++$ is a least element of A , which contradicts our supposition that A has no least element.

Therefore no element of A is $\leq n++$, which means $P(n++)$ is true.

Conclusion :

Using induction, we have proven that **if** A exists as defined earlier, $P(n)$ holds for every $n \in A$.

Since $A \neq \emptyset$, let us take some $a \in A$.

Applying $P(a)$: no element of A is $\leq a$.

But $a \in A$ and $a \leq a$, a contradiction.

Hence, either A is non-empty, or the supposition that A has no least elements was false, and A *does* have a least element, i.e. every non-empty subset of $\mathbb{N}$ has a least element, which means $\mathbb{N}$ is a well-ordered set.

From the above statements, we can see that, taking axioms P1-P4, it is not possible to derive induction from additionally assuming well-ordering without adding modifications to the original statement. However, even if we additionally assume only induction and no other modifications, we are able to derive the well-ordering principle.

NOTE: F1-F3 are not modifications to the statement, since they can also be proven using P1-P4 and induction (given at the end of these notes), and are implicitly assumed in this case to reduce complexity of the proof.

I.e.,

$P1-P4 + \text{Well-ordering} \not\Rightarrow \text{Induction}$

$P1-P4 + P5 \text{ (Induction)} \Rightarrow \text{Well-ordering}$

$P1-P4 + (A) + (W) + (S) + (O) \Rightarrow \text{Induction}$

From this, we can show why **Induction is considered a foundational axiom about how $\mathbb{N}$ is constructed using Peano Axioms**, while **Well-Ordering is just a principle about a property that it has**.

NOTE : We can build a valid alternate system for $\mathbb{N}$ which is based on the primitive order instead of on Peano's Axioms. In this system, the well-ordering principle would be primary, and we could imply induction from it. Although this system is equivalent to the system based on Peano's Axioms, we will use the latter in this course, and consider induction to be the foundational axiom from which well-ordering can be implied.

Statement. *Let $(M, <)$ be a linearly ordered set satisfying the following axioms:*

- (O1) $<$ is a strict linear order on a nonempty set M .*
- (O2) Every nonempty subset of M has a least element.*
- (O3) M has no greatest element.*
- (O4) Every nonempty subset of M that is bounded above has a greatest element.*

Define:

$0 :=$ the least element of M

$n++ :=$ the least element of $\{x \in M : n < x\}$

The first is well-defined by (O2) applied to M , which is nonempty; the second by (O3), which guarantees that $\{x : n < x\}$ is nonempty, since if the set is empty, n would become the greatest element of M , together with (O2), due to which it has to have a least element.

Then the principle of mathematical induction holds: if $S \subseteq M$ satisfies $0 \in S$ and $n \in S \Rightarrow n++ \in S$, then $S = M$.

First, let us understand *what* S is. **Informally** speaking, elements of the set S can be considered to be the elements for which a property 'P' is considered true. Therefore, if an arbitrary $s \in M$ is $\in S$, it can be considered the same as saying $P(s) = \text{true}$.

Since we haven't explicitly defined what a property would be in this version of $\mathbb{N}$, we cannot directly use P . Hence, any mention of this property P in this proof is purely for the purpose of clarification, and **is not formal**.

Thus, the induction statement can be "rewritten" as : if a property P on the elements of M (i.e., **if a set $S \subseteq M$ satisfies** $P(0) = \text{true}$ (i.e. $0 \in S$) **and** $P(m) = \text{true} \Rightarrow P(m++) = \text{true}$ (i.e. $n \in S \Rightarrow n++ \in S$), **then** $P(m) = \text{true}$ for all $m \in M$ (i.e., $S = M$).

To connect this statement back to our area of interest (i.e., $\mathbb{N}$), while M can be any set of elements for which the definitions given in the statement hold, the set of natural numbers $\mathbb{N}$ is the most intuitive example of such a set. Therefore, all the statements made in the below proof which apply to M , apply to $\mathbb{N}$ as well.

Proof. Let us begin the actual proof :

Step 1. Suppose $M \setminus S \neq \emptyset$ (essentially, assume the set of (elements for which $P = \text{false}$) $\neq \emptyset$).

Step 2. By (O2), $M \setminus S$ has a least element m i.e. $m \in M \setminus S$.

Step 3. Since $0 \in S$, $0 \notin M \setminus S$, hence, $m \neq 0$.

Step 4. Since $m \in S$, and $S \subset M$, $m \in M$ and therefore $0 < m$, since 0 is the least element of M .

Step 5. Hence, we can define a set

$$L = \{x \in M : x < m\}$$

i.e. the set of all elements in M that are $< m$.

This set is nonempty, since 0 is necessarily $\in L$, and L is bounded above by m (i.e. m is greater than all elements of L).

Step 6. By (O4), L has a greatest element p . $p \in L$ gives $p < m$.

Step 7. We claim $p++ = m$. By the second definition in the statement, this can be rewritten as claiming that m is the least element of $\{x \in M : p < x\}$. Let us prove by contradiction.

Suppose that there is some k that satisfies $p < k < m$, i.e. some k that is the least element of the above set instead of m .

Then $k \in L$ and $p < k$, contradicting the maximality of p in L . So, we get a contradiction, and therefore, k cannot exist. This means m must be the least element of above set, i.e. $m = p++$.

Step 8. Since $p \in L$, $p \in M$.

Step 9. Since $p < m$ and m is the least element of $M \setminus S$, $p \notin M \setminus S$.

Step 10. Thus, we get $p \in S$. This means that $p++ = m \in S$, according to the inductive hypothesis, and therefore, $m \notin M \setminus S$.

This contradicts our initial supposition that $M \setminus S \neq \emptyset$. Hence $M \setminus S = \emptyset$, i.e. $S = M$.

From this, we can say that, with this alternate system of building $\mathbb{N}$, well-ordering serves as a foundational axiom, while induction is a property of $\mathbb{N}$.